

Pei (Patrick) Chen | 陈沛

[Personal Website](#) | [Google Scholar](#)

chenpei.net@gmail.com | +1 979-739-4507

Professional Summary

Applied Scientist at Amazon building the **agentic capability stack** around large MoE models (400B+): **agentic data synthesis**, **verifiable RL**, **harness/memory/personalization**, **long-context**, and **rubric-based evaluation** for customer-facing agents. **Mid-training** builds the substrate — long-context from 8K to 128K (ABF/YaRN; **training-free to 1M**) and fundamental agent capability synthesized from **real, executable agent environments**. **Post-training** and RL shape behavior through **verifiable rewards and graders** — **multi-turn robustness**, **tool-use**, **personalization**, and faithfulness — closed by a **production data flywheel** that compounds live-traffic failures into continuous improvement via standardized data generation and automated critics.

Ph.D. in Computer Science with **20+ publications** (8 first-authored) at **NeurIPS (Spotlight)**, ACL, EMNLP, and NAACL, spanning long-context and agentic capabilities, multi-turn modeling, and personalization; representative first-author work: **HYTREL** (NeurIPS Spotlight), which reworked self-attention into a set-attention architecture for tabular LMs, and **CoMM**, an early multi-agent reasoning harness for complex problem solving; one US patent, Area Chair at ARR.

Experience

Amazon

2024.1 – Present

Applied Scientist (full-time)

Bay Area, CA

- **Rufus Foundation Modeling (startup-like) — Model Mid/Post-training & Evaluation:** Led two tracks on **400B+ MoE** backbones for customer-facing shopping agents: **mid-training** for the general capability substrate, and **post-training, RL, and rubric-based evaluation** for model behavior.
 - **Agent training: mid-training** on synthesized agent corpora (**103B tokens / 76.5K APIs**, 1:1:1 agent-code-text mix) builds the capability substrate: BFCL-v2 59.6 → 70.8, BFCL-v3 35.8 → 50.6 (**Hephaestus**, NAACL 2025); **RL** then shapes behavior on tasks synthesized from the agent's own interaction with the environment — gold-traced verified trajectories plus failure-mined tasks — lifting τ^2 -Bench Retail Pass¹ to 82.5 on a 30BA3B-model, surpassing GPT-5.2 and ranked #2 at the time of submission.
 - **Long-context: mid-training** to extend long-context from **8K to 128K** tokens through long-context data curation and positional scaling (**ABF, YaRN**); training-free to **1M**, with a **progressive-disclosure memory harness** surfacing only the context a turn needs — matching GPT-OSS-120B, a leading open model then.
 - **Personalization:** translated ambiguous personalization goals into **verifiable rubrics and reward signals**, raising rubric-human alignment from **40% to 78%** over five iterations, then used the resulting reward design to drive **GRPO**, lifting personalization helpfulness from **84% to 93%**, and enabling **production dial-up**.
 - **Faithfulness & grounding:** increased faithfulness from **67% to 90%** on customer-facing **RAG** evaluations by building **synthetic IFT + DPO** pipelines for hallucination reduction, closing hallucination as a launch blocker.
- **Customer Service — Post-training Data Infrastructure & Flywheel:** Owned the end-to-end post-training data flywheel for a multi-agent LLM system, productionizing scalable annotation, quality control, data selection, and versioning pipelines for SFT/DPO/GRPO, retrieval, and evaluation data. Developed diversity- and difficulty-aware data selection strategies that improved policy prediction accuracy from **86% to 93%** and response quality from **88% to 91%**, while reducing training time by **75%** for policy prediction training (**24h to 6h**) and **80%** for response generation training (**10d to 2d**).
- **Research:** long-context modeling and evaluation; mid-training for fundamental agent capabilities; verifiable **RL (GRPO)** for agentic tool use, multi-turn robustness, and personalization; mid/post-training data synthesis. **10+** publications; **primary mentor** for two research interns.

Amazon Web Services (AWS)

2023.5 – 2023.8

Applied Scientist (Intern)

Santa Clara, CA

- Developed a **collaborative multi-agent** prompting framework (**CoMM**) achieving SOTA on complex science

benchmarks in zero-shot and few-shot settings — published at NAACL 2024.

Amazon Web Services (AWS)

2022.6 – 2023.1

Applied Scientist (Intern)

Santa Clara, CA & remote

- Developed [HYTREL](#), a **hypergraph-based tabular LM** capturing structural dependencies, achieving SOTA on tabular representation learning — published as **NeurIPS 2023 Spotlight (top 5%)**.

Tencent AI Lab

2021.6 – 2021.8

NLP Researcher (Intern)

remote (College Station, TX)

- Built a benchmark for zero-shot knowledge base completion across diverse KB sources — published at ICDM 2022 KG Workshop.

National Lab of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences

2018.1 – 2019.7

Research Engineer (Intern & Full-time)

Beijing, China

- Built information extraction and evaluation systems for financial event extraction and causality detection.

Selected Publications

LLM Training & Alignment — *Data-centric and structure-aware approaches to LLM training and alignment.*

- **Pei Chen**, Soumajyoti Sarkar, Leonard Lausen, et al. [“HYTREL: Hypergraph-enhanced Tabular Data Representation Learning.”](#) **NeurIPS 2023, Spotlight (5%), first work to use set-attention for tabular LMs.**
- Ming Li, **Pei Chen**, Chenguang Wang, et al. [“Mosaic-IT: Cost-Free Compositional Data Synthesis for Instruction Tuning.”](#) ACL 2025, Findings.
- Hongye Jin, **Pei Chen**, Jingfeng Yang, et al. [“END: Early Noise Dropping for Efficient and Effective Context Denoising.”](#) pre-print.
- Wangtao Sun, Haotian Xu, Xuanqing Yu, **Pei Chen**, et al. [“ItD: Large Language Models Can Teach Themselves Induction through Deduction.”](#) ACL-2024.
- Zhaoxuan Tan, Zheng Li, Tianyi Liu, Haodong Wang, **Pei Chen**, et al. [“Aligning Large Language Models with Implicit Preferences from User-Generated Content.”](#) ACL 2025.

Agent — *Multi-agent reasoning, agent-oriented pre-training, verifiable tool-use training, and agent self-evolution.*

- **Pei Chen**, Shuai Zhang, Boran Han. [“CoMM: Collaborative Multi-Agent, Multi-Reasoning-Path Prompting for Complex Problem Solving.”](#) NAACL 2024, **early work in demonstrating multi-agent systems for complex LLM reasoning.**
- Chen Ling, **Pei Chen**, Albert Guan, et al. [“PACE: Two-Timescale Self-Evolution for Small Language Model Agents.”](#) pre-print.
- Yuchen Zhuang, Jingfeng Yang, Haoming Jiang, Xin Liu, **Pei Chen**, et al. [“Hephaestus: Improving Fundamental Agent Capabilities of Large Language Models through Continual Pre-Training.”](#) NAACL 2025.
- Hy Dang, Tianyi Liu, Zhuofeng Wu, Jingfeng Yang, **Pei Chen**, et al. [“Improving Large Language Models Function Calling and Interpretability via Guided-Structured Templates.”](#) EMNLP 2025.
- Ziyi Wang, Yuxuan Lu, Yimeng Zhang, **Pei Chen**, et al. [“Trajectory2Task: Training Robust Tool-Calling Agents with Synthesized Yet Verifiable Data for Complex User Intents.”](#) accepted to ACL 2026.

Model Behavior — Evaluation, Multi-turn, Memory & Personalization — *Measuring and training conversational behavior across long-context, multi-turn, retrieval, and memory.*

- **Pei Chen**, Hongye Jin, Cheng-Che Lee, et al. [“LongLeader: A Comprehensive Leaderboard for Large Language Models in Long-context Scenarios.”](#) NAACL 2025.
- Ming Li, **Pei Chen**, Zhenhao Zhang, et al. [“Mitigating Lost in Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards.”](#) Accepted to ACL 2026.
- Yichuan Li, Xinyang Zhang, Chenwei Zhang, Mao Li, **Pei Chen**, et al. [“ALERT: An LLM-powered Benchmark for Automatic Evaluation of Recommendation Explanations.”](#) NAACL 2025.

- Fengran Mo, Yifan Gao, Chuan Meng, Xin Liu, **Pei Chen**, et al. “UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations.” ACL 2025.
- Fengran Mo, Yifan Gao, Zhuofeng Wu, Xin Liu, **Pei Chen**, et al. “Leveraging historical information to boost retrieval-augmented generation in conversations.” Information Processing & Management, March 2026.
- Zhaoxuan Tan, Zixuan Zhang, Haoyang Wen, Zheng Li, **Pei Chen**, et al. “Instant Personalized Large Language Model Adaptation via Hypernetwork.”, accepted to ACL 2026.
- Yizhuo Chen, Xin Liu, Ruijie Wang, Zheng Li, **Pei Chen**, et al. “POPI: Personalizing LLMs via Optimized Natural Language Preference Inference.” pre-print.

Information Extraction — *Graph-based modeling of non-sequential dependencies in structured extraction (Ph.D. work).*

- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. “Explicitly Capturing Relations between Entity Mentions via Graph Neural Networks for Domain-specific Named Entity Recognition.” ACL-IJCNLP 2021.
- **Pei Chen**, Haotian Xu, Cheng Zhang, and Ruihong Huang. “Crossroads, Buildings and Neighborhoods: a Dataset for Fine-grained Location Recognition.” NAACL 2022.
- **Pei Chen**, Hang Yang, Kang Liu, et al. “Reconstructing Event Regions for Event Extraction via Graph Attention Networks.” AACL-IJCNLP 2020.
- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. “Domain-specific Named Entity Recognition via Graph Neural Networks.” US Patent 12,530,529, 2026.

Education

Texas A&M University	2019.8 - 2024.5
<i>Ph.D. in Computer Science</i>	<i>College Station, TX</i>

- **GPA:** 4.0/4.0
- **Research Directions:** Information extraction and structured NLP (NER, event extraction, knowledge base completion); later extended to LLMs and agentic systems.
- **Dissertation:** Connecting the Dots: Improving Information Extraction by Modeling the Non-Sequential Dependencies of Entity Mentions.
- **Teaching assistant:** CSCE 636 Deep Learning and CSCE 313 Computer Systems.

Southwestern University of Finance and Economics	2016.9 – 2018.6
<i>Master’s Degree in Finance</i>	<i>Chengdu, China</i>

- **GPA:** 3.9/5.0, ranking 1/178.
- **Awards:** 2017 National Scholarship for Graduate Student.

National University of Defense Technology	2010.9 – 2014.6
<i>B.Engr. in Simulation Engineering</i>	<i>Changsha, China</i>

- **GPA:** 88.61/100, ranking 1/45.
- **Thesis:** Analyze and reconstruct the multi-resolution modeling technology of a simulation system written by millions of lines of C++ code.

Professional Service

- 2026: **Area Chair for ARR**, Program Committee/Reviewer for ARR, ICLR 2026, ICML 2026.
- 2025: Program Committee/Reviewer for ARR, COLING 2025, ICLR-2025, NeurIPS 2025
- 2024: Program Committee/Reviewer for ARR, NeurIPS 2024, TPAMI
- 2023: Program Committee/Reviewer for ACL, EMNLP, ARR
- 2022: Program Committee/Reviewer for EMNLP, ARR, NLPCC
- 2021: Program Committee/Reviewer for EMNLP, ARR, NLPCC

Skills

- **Mid-training:** long-context data curation, positional scaling (ABF, YaRN), agent-environment data synthesis
- **Post-training & RL:** SFT, DPO, GRPO/DAPO, reward design
- **Evaluation Methodology:** verifiable rubric design, human alignment calibration (iterative refinement), LLM-as-judge validation, synthetic eval data generation
- **Agent environments & Harness:** executable environment synthesis for verifiable agent data, progressive-disclosure memory system, multi-agent orchestration (CoMM)
- **Training frameworks & Programming:** Megatron, Verl, Slime; Python, PyTorch, SQL, C/C++, Git, Docker.